NIMCET 2016

Mathematics:

1. The number of 5 people groups that can be selected from 9 people when two particular persons are not to be in the same group is
(A) 126 (B) 35
(C) 91 (D) 252

2. The solution set of equation $\log_x 2\ \log_{2x} 2 = \log_{4x} 2$ is
(A) $\left\{2^{-\sqrt{2}}, 2^{\sqrt{2}}\right\}$ (B) $\left\{\frac{1}{2}, 2\right\}$
(C) $\left\{\frac{1}{4}, 2^2\right\}$ (D) $\left\{\frac{1}{4}, 2\right\}$

3. If a twelve sides regular polygon is inscribed in a circle of radius 3 centimeters, then the length of each side of the polygon is
(A) 3 (B) $18 - 9\sqrt{3}$
(C) $18 + 9\sqrt{3}$ (D) $9(1 - \sqrt{3})$

4. If C is the mid-point of AB and P is point outside AB, then
(A) $PA + PB = 2PC$ (B) $PA + PB = PC$
(C) $PA + PB + 2PC = 0$ (D) $PA + PB + PC = 0$

5. The average marks of boys in class is 52 and that of girls to 42. The average marks of boys and girls combined is 50, then percentage of boys in the class is
(A) 80% (B) 60%
(C) 40% (D) 20%

6. A box contains 2 blue caps, 4 red caps, 5 green caps and 1 yellow cap. If four caps are picked at random, then the probability that none of them is green, is
(A) $\frac{7}{99}$ (B) $\frac{7}{12}$
(C) $\frac{5}{99}$ (D) $\frac{5}{12}$

7. The line $3x + 5y = k$ touches the ellipse $16x^2 + 25y^2 = 400$ if k is
(A) $\pm\sqrt{5}$ (B) $\pm\sqrt{15}$
(C) ± 25 (D) $\pm\sqrt{35}$

8. If $X = \{4^n - 3n - 1, n \in N\}$ and $Y = \{9n - 9, n \in N\}$, then $X \cup Y$ is equal to
(A) Y (B) X
(C) N (D) None of these

9. $\int \left\{\frac{(\log x - 1)}{1 + (\log x)^2}\right\}^2 dx$ is equal tp
(A) $\frac{xe^x}{1+x^2} + C$ (B) $\frac{x}{(\log x)^2 + 1} + C$
(C) $\frac{\log x}{(\log x)^2 + 1} + C$ (D) $\frac{x}{x^2 + 1} + C$

10. The volume of the parallelepiped determined by $U = \hat{\imath} + 2\hat{\jmath} - \hat{k}, V = -2\hat{\imath} + 3\hat{k}$ and $W = 7\hat{\jmath} - 4\hat{k}$ is
(A) 21 (B) 22
(C) 23 (D) 24

11. The vector perpendicular to the plane passing through $(1, -1, 0), (-2, 1, -1)$ and $(-1, 1, 2)$ is
(A) $6\hat{\imath} + 6\hat{k}$ (B) $6\hat{\imath} + 7\hat{k}$
(C) $7\hat{\imath} + 6\hat{k}$ (D) $7\hat{\imath} + 8\hat{k}$

12. The equation of a circle with diameters are $2x - 3y + 12 = 0$ and $x + 4y - 5 = 0$ and area of 154 sq units is
(A) $x^2 + y^2 - 6x + 4y - 36 = 0$
(B) $x^2 + y^2 + 6x - 4y - 36 = 0$
(C) $x^2 + y^2 - 6x - 4y + 25 = 0$
(D) None of the above

13. $\int \frac{x^2 - 1}{x^3\sqrt{2x^4 - 2x^2 + 1}} dx$ is equal to
(A) $\frac{\sqrt{2x^4 - 2x^2 + 1}}{x^2} + C$ (B) $\frac{\sqrt{2x^4 - 2x^2 + 1}}{x^3} + C$
(C) $\frac{\sqrt{2x^4 - 2x^2 + 1}}{x} + C$ (D) $\frac{\sqrt{2x^4 - 2x^2 + 1}}{2x^2} + C$

14. If a, b and $a + b$ are vectors of magnitude α, then the magnitude of vector $a - b$ is
(A) $\sqrt{2}\alpha$ (B) $\sqrt{3}\alpha$
(C) $2\ \alpha$ (D) 3α

15. Which of the following statements is FALSE?
(A) $2 \in A \cup B$ implies that if $2 \notin A$, then $2 \in B$
(B) $[2, 3] \subseteq A$ implies that $2 \subseteq A$ and $[2, 3] \subseteq B$

(C) $A \cup B \supseteq [2,3]$ implies that $[2,3] \subseteq$ A and $[2,3] \subseteq B$

(D) $[2] \in$ A nad $[3] \in$ A implies that $[2,3] \subseteq$ A

16. If $2x^2 + 7xy + 3y^2 + 8x + 14y + \lambda = 0$ represents pair of straight lines, the value of λ is

(A) 2 (B) 4

(C) 6 (D) 8

17. The area of the region bounded by the lines $y = |x - y|$ and $y = 3 - |x|$ is

(A) 3 sq units (B) 4 sq units

(C) 6 sq units (D) 2 sq units

18. In a triangle $ABC, a = 4, b = 3, \angle BAC = 60°$, then the equation for which c is the root, is

(A) $c^2 + 3c + 7 = 0$ (B) $c^2 + 3c - 7 = 0$

(C) $c^2 - 3c + 7 = 0$ (D) $c^2 - 3c - 7 = 0$

19. If $\cos\theta = \frac{5}{13}, \frac{3\pi}{2} < \theta < 2\pi$, then $\tan 2\theta$ is

(A) $\frac{-120}{119}$ (B) $\frac{-120}{169}$

(C) $\frac{119}{116}$ (D) $\frac{120}{119}$

20. An expression has 10 equally likely outcomes. Let A and B be two non-empty events of the experiment. If consists of 4 outcomes, then number of outcomes that B must have so that A and B are independent, is

(A) 2, 4 or 8 (B) 3, 6 or 9

(C) 4 or 8 (D) 5 or 10

21. Let a, b and c be three non-zero vectors, no two of which are collinear. If the vector $a + 2b$ is collinear with c and $b + 3c$ is collinear with a, then $a + 2b + 6c$ is equal to

(A) λa (B) λb

(C) λc (D) 0

22. The value of a, for which the sum of the square of the roots of the equation $x^2 - (a-2)x - (a+1) = 0$, assumes the least value is

(A) 3 (B) 2

(C) 0 (D) 1

23. For any two events A and B, the probability that atleast one of them occur is 0.6. If A and B occurs simultaneously with a probability 0.3, then $P(A') + P(B')$ is

(A) 0.9 (B) 1.15

(C) 1.1 (D) 1.0

24. Two finite sets A and B are having m and n elements. The total number of subsets of the first set is 56 more than the total number of subsets of the second set. The value of m and n are

(A) 7, 6 (B) 6, 3

(C) 5, 3 (D) 8, 7

25. The probability that A speaks truth is 4/5 while this probability for B is 3/4. The probability that they contradict each other then when asked to speak on a faci is

(A) 3/20 (B) 1/5

(C) 7/20 (D) 4/5

26. The sum of the expression $\frac{1}{\sqrt{1}+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + \frac{1}{\sqrt{3}+\sqrt{4}} + \cdots + \frac{1}{\sqrt{80}+\sqrt{81}}$ is

(A) 7 (B) 8

(C) 9 (D) 10

27. Consider the function f defined by $f(x) = \begin{cases} x^2 - 1, & x < 3 \\ 2ax, & x \geq 3 \end{cases}$ for all real number x. If f is continuous at $x = 3$, then value of a is

(A) 8 (B) $\frac{3}{4}$

(C) $\frac{1}{8}$ (D) $\frac{4}{3}$

28. Three houses are available in a locality. Three persons apply for the houses. Each applies for one house without consulting others. The probability that all the three apply for the same house is

(A) $\frac{8}{9}$ (B) $\frac{7}{9}$

(C) $\frac{2}{9}$ (D) $\frac{1}{9}$

29. Five horses are in a race. Mr. A selects two of the horses at random and bets on them. The probability that Mr. A selected the winning horse is

(A) $\frac{3}{5}$ (B) $\frac{1}{5}$

(C) $\frac{2}{5}$ (D) $\frac{4}{5}$

30. If $3^{x} = 4^{x-1}$, then x is equal to

(A) $\frac{2-\log_3 2}{2\log_3 2-1}$ (B) $\frac{2}{2\log_3 2-1}$

(C) $\frac{2-\log_3 2}{2\log_3 2+1}$ (D) $\frac{2\log_3 2}{2\log_3 3-1}$

31. The matrix A has x rows and $(x+5)$columns and the matrix D has y rows and $(11-y)$ columns. If both the matrices AB and BA exist, then the value of x and y are

(A) 8, 3 (B) 3, 5

(C) 3, 8 (D) 8, 5

32. A circus artist is climbing a 20 m long rope, which is tightly stretched and tied from the top of a vertical pole to the ground. Find the height of the pole, if the angle made by the rope with the ground level is 30°.

(A) 10 m (B) 20 m

(C) 30 m (D) 40 m

33. There are n equally speed points 1, 2, ..., n marked on the circumference of a circle. If the point 15 is directly opposite of the point 49, then the total number of points is

(A) 50 (B) 68

(C) 66 (D) 70

34. Let $S = \{1, 2, \ldots, n\}$. The number of possible pairs of form (A, B) with $A \subseteq B$ for subsets A, B of S is

(A) 2^n (B) 3^n

(C) nl (D) $\sum_{k=1}^{n} \binom{n}{k}\binom{n}{n-k}$

35. Sum of the roots of the equations $4^x - 3(2^{x+3}) + 128 = 0$ is

(A) 5 (B) 6

(C) 7 (D) 8

36. If the sum of the slope of the lines given by $x^2 - 2cxy - 7y^2 = 0$ is four times their product, then the value of c is

(A) 1 (B) -1

(C) -2 (D) 2

37. The system of equations

$$x + y + 2z = a$$
$$x + z = b$$
$$2x + y + 3z = c$$

has a solution if

(A) $b = c$ (B) $c = a + b$

(C) $c = a + 2b$ (D) $a = b = c$

38. Let $f(x) = x^2 - bx + c$, b is an odd positive integer. If $f(x) = 0$ has two prime number as roots and $b + c = 35$, then the global minimum value of $f(x)$ is

(A) $\frac{-183}{4}$ (B) $\frac{173}{16}$

(C) $\frac{-81}{4}$ (D) $\frac{17}{2}$

39. The vertex of the parabola whose focus is $(-1, 1)$ and directrix is $4x + 3y - 24 = 0$ is

(A) $\left(0, \frac{3}{2}\right)$ (B) $\left(0, \frac{5}{2}\right)$

(C) $\left(1, \frac{3}{2}\right)$ (D) $\left(1, \frac{5}{2}\right)$

40. The number of points in $(-\infty, \infty)$, for which $x^2 - x\sin x - \cos x = 0$ is

(A) 6 (B) 4

(C) 2 (D) 0

41. There are 4 books on fairy tales, 5 novels and 3 plays. In how many ways can they be arranged in the order, books on fairy tales, novels and then plays so that the books of the same category are put together?

(A) 17280 (B) 103680

(C) 51840 (D) 360

42. Suppose population A has 100 observations 101, 102, ..., 200 and another population B has 100 observation 151, 152, ..., 250. If V_A and V_B represent variance of the two population respectively, then $\frac{V_A}{V_B}$ is

(A) $\frac{9}{4}$ (B) $\frac{4}{9}$

(C) 1 (D) $\frac{2}{3}$

43. If a, b are vectors such that $|a + b| = \sqrt{29}$ and $a \times (2\hat{\imath} + 3\hat{\jmath} + 4\hat{k}) = (2\hat{\imath} + 3\hat{\jmath} + 4\hat{k}) \times b$, then a possible value of $(a + b).(-7\hat{\imath} + 2\hat{\jmath} + 3\hat{k})$ is

(A) 0 (B) 3

(C) 4 (D) 8

44. Let $x_1, x_2, \ldots, x_n$ be n observation such that $\sum x_i^2 = 400$ and $\sum x = 80$. Then a possible value of n among the following is

(A) 10
(B) 15
(C) 20
(D) 8

45. Area of the greatest rectangle that can be inscribed in the ellipse is
(A) $\sqrt{ab}$
(B) $2ab$
(C) ab
(D) $\frac{a}{b}$

46. Two common tangents to the circle $x^2 + y^2 = 2a^2$ and parabola $y^2 = 8ax$ are
(A) $x = \pm(y + 2a)$
(B) $y = \pm(x + 2a)$
(C) $x = \pm(y + a)$
(D) $y = \pm(x + a)$

47. If $a_1, a_2, \dots, a_n$ are in AP and $a_1 = 0$, then the value of $\left(\frac{a_3}{a_2} + \frac{a_4}{a_3} + \cdots + \frac{a_n}{a_{n-1}}\right) - a_2\left(\frac{1}{a_2} + \frac{1}{a_3} + \cdots + \frac{1}{a_{n-2}}\right)$ is equal to
(A) $(n-2) + \frac{1}{(n-2)}$
(B) $\frac{1}{n-2}$
(C) $n - 2$
(D) $n - \frac{1}{n-2}$

48. The value of $\cos 20^\circ + \cos 100^\circ + \cos 140^\circ$ is
(A) 0
(B) $\frac{1}{\sqrt{2}}$
(C) $\frac{1}{2}$
(D) 1

49. The permutations of (a, b, c, d, e, f, g) are listed in lexicographic order. Which of the following permutations are just before and just after the permutation bacdefg?
(A) agfedbc and bacdfge
(B) agfedcb and badcefg
(C) agfebcd and bacedgf
(D) agfedcb and bacdegf

50. The foci the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ and the hyperbola $\frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{25}$ coincide. Then the value of b^2 is
(A) 5
(B) 7
(C) 9
(D) 1

Reasoning/Aptitude:

Directions [Q. Nos. 51-52] are based on the following: A boys is asked to put in a basket one mango when ordered 'One', one orange when ordered 'Two'. One apple when ordered 'Three' and is asked to take out from the basket one mango and an orange when ordered 'Four'. A Sequence of order is given as
1233214231422331 4113234

51. How many total fruits will be in the basket at the end of the above order sequence?
(A) 9
(B) 8
(C) 11
(D) 10

52. How many total oranges were in the basket at the end of the above sequence?
(A) 1
(B) 4
(C) 3
(D) 2

Directions [Q. Nos. 53 to 56] are based on the following: Eight Friends J, K, L, M, N, N, O, P and Q live on eight different floors of a building but not necessarily in same order. The lower most floor of the building is numbered one, the above that numbered two and soon until the loop most floor is numbered eight.

- J lives on floor numbered six.
- Only one person lives between J and L.
- O lives on the floor immediately below L.
- Only one person lives between O and P.
- O lives above P.
- K lives on an even numbered floor but not on the floor numbered two.
- Two persons live between K and Q.
- Q does not live on the lower most floor.
- N lives on one of the floor above Q.

53. If P and L Interchange their places, who will live between P and M?
(A) O
(B) L
(C) J
(D) No one

54. Which of the following is true about M?
(A) K lives immediately above M
(B) Only two people live between M and Q
(C) M lives on an odd numbered floor
(D) M lives on the lower most floor

55. Who amongst the following lives on the floor numbered eight?

(A) P
(B) O
(C) K
(D) Cannot be determined

56. Three of the following four are alike in a certain way based on the given arrangement and thus form a group. Which of the following does not belong to the group?

(A) PL (B) MQ
(C) LN (D) OM

Directions [Q. Nos. 57 to 59] are based on the following: A team must be selected from the ten probable players A, B, C, D, E, F, G, H, I and J. Of these, A, C, E and J are forwards B, G and H are point guards and D, F and I are defender.

- The team must have at least one forward, one point guard and one defender.
- If the team include J, it must also include F.
- The team must include E or B, but not both.
- If the team include G, it must also include F.
- The team must include exactly one among C, G and I.
- C and F cannot be members of the same team.
- D and H cannot be members of the same team.
- The team must include both A and D or neither of them.
- There is no restriction on the number of members in the team.

57. What would be the size of the largest possible team?

(A) 7 (B) 6
(C) 5 (D) 4

58. Which of the following cannot be included in a team of size 6?

(A) A (B) H
(C) J (D) E

59. What could be the maximum size of the team that includes G?

(A) 4 (B) 5
(C) 6 (D) More than 6

60. A family has several children. Each boy in his family has as many sisters as brothers, but each girl has twice as many brothers as sister. How many brothers and sister are there?

(A) 1 and 2 (B) 3 and 4
(C) 6 and 3 (D) 4 and 3

61. In a certain code language '134' means 'good and tasty', '478' means 'see good picture' and '729' means 'picture are faint'. Which of the following numerical symbols stands for see?

(A) 2 (B) 7
(C) 8 (D) 1

62. If the English word 'EXAMINATION' is coded as 56149512965, then word 'GOVERNMENT' can be coded as

(A) 7655955552 (B) 7645954452
(C) 7645954552 (D) 7644956552

63. Fact 1: Most stuffed toys are stuffed with beans.
Fact 2: There are stuffed bears and stuffed tigers.
Fact 3: Some chairs are stuffed with beans.
If the above statements are fact. Which of the following statements, must also be fact?
1. Only children's chairs are stuffed with beans.
2. All stuffed tigers are stuffed with beans.
3. Stuffed monkeys are not stuffed with beans.

(A) 1 is a fact
(B) Only 2 is a fact
(C) Both 2 and 3 are facts
(D) None of the statements 1, 2, 3 are true

64. If all the 6's are replaced by 9's then the algebraic sum of all the numbers from 1 to 100 (both inclusive) varies by

(A) 333 (B) 300
(C) 279 (D) 330

65. 405 sweets were distributed equally among a group of children such that the number of sweet received by each child is one fifth of the number of children. The number of children in the group is

(A) 45 (B) 9
(C) 21 (D) 15

66. The number of common terms in the two sequence 17, 21, 25, ………, 817 and 16, 21, 26, ………, 851 is
(A) 28 (B) 39
(C) 40 (D) 87

67. Unscramble the letter in the following words and find the odd one.
(A) ONGEAR (B) NOONI
(C) ALPEP (D) AUVAG

68. A bus starts from its depot filled to seating capacity. It stops at a point A where 1/6th the of passengers alight and 10 board the bus. At point B, 1/5th of the passengers alight and boards the bus. At point C which is the last stop, all the 55 passengers alight. The capacity of the bus is
(A) 96 (B) 99
(C) 66 (D) 90

69. Kha-kha is an obscure island which is inhabited by two types of people; the 'Yes' type and the 'No' type. Native of types 'Yes' ask only questions the right answer in which is 'Yes' While those of type 'No' ask only questions the right answer to which is 'No'. For example, the 'Yes' type will ask questions like "Is 2 plus 2 equals to 4?" While the "No' type will ask question like "Is 2 plus 2 equals to 5?". The following question is based on your visit to the island Kha-kha. Kevin and Kumar are brothers from the island. Kumar asks you, Is at least one of us is of type 'No'? You can conclude that
(A) Kevin is 'No', Kumar is 'Yes'.
(B) Both are 'Yes'.
(C) Kevin is 'Yes', Kumar is 'No'
(D) Both are 'No'

70. Raman was born on march 5, 1970. Lakshman was born 25 days before Raman. The year when they took birth, the Republic Day fell on Monday. What is the day of birth of Lakshman
(A) Sunday (B) Monday
(C) Wednesday (D) Saturday

71. P, Q, R and S are four logical statements such that if P is true, then Q is true; if Q is true, then R is true; and if S is true, then at least one of Q and R is false. Then it follows that
(A) If S is false, then both Q and R true
(B) If at least one of Q and R is true, then S is false
(C) If P is true, then S is false
(D) If Q is true, then S is true

72. A, B, C, D, E, F and G are members of a family consisting of four adults, three children, three males and four females. Out of the children F and G are girls. A and D are brothers and A is doctor, E is an engineer married to one of the brothers and has two children. B is married to D and G is their child. Who is C?
(A) E's daughter (B) A's son
(C) G's brother (D) F's father

73. A clock is set right at 5 am the clock losses 16 minutes in 24 hours. What will be the correct time when the clock indicates 10 pm on the 3rd day?
(A) 11 pm (B) 10:45 pm
(C) 11:15 pm (D) 12 pm

74. The day after the day after tomorrow is four days after Monday. What day is it today?
(A) Monday (B) Tuesday
(C) Wednesday (D) Thursday

Direction [Q. Nos. 75 to 77]

- There are six houses P, Q, R, S, T and U, three on either side of a road.
- The houses are of different colours – red, blue, green, orange, yellow and white.
- All the houses are of different heights.
- T, the tallest house is exactly opposite to the red coloured house.
- The shortest house is exactly opposite to the green coloured house.
- U, the orange coloured house is located between P and S
- R, the yellow coloured house is exactly opposite to P.
- Q, the green coloured house is exactly opposite to U.
- P, the white coloured house is taller than R, but shorter than S and Q.

75. Which of the second largest house?
(A) Q (B) R
(C) S (D) Cannot be determined

76. Which of the second shortest house?

(A) P (B) R
(C) S (D) Cannot be determined

77. What is the colour of the tallest house?

(A) Red (B) Blue
(C) Green (D) Yellow

78. How many pairs of letters are there in the word NECESSARY which have as many letters between them in the word as there are between them in the alphabet and in the same order.

(A) One (B) Two
(C) Three (D) Four

79. What is missing number in the series, 4, 7, 11, 18, 29, 47, ………, 123, 199?

(A) 76 (B) 77
(C) 86 (D) 87

Directions [Q. Nos. 80 to 82]

- A, B, C, D, E and F are six members in a family in which there are two married couples.
- D is brother of F.
- Both D and F are lighter than B.
- B is mother of D and lighter than E.
- C a lady, is neither heaviest nor lightest in the family.
- E is lighter than C
- The grandfather in the family is the heaviest.

80. Which of the following is a pair of married couples.

(A) AB (B) BC
(C) AD (D) BE

81. Who among the following will be in the second place, if all the members in the family are arranged in a descending order of their weights?

(A) C (B) A
(C) D (D) Data inadequate

82. How is C related to D?

(A) Grandmother (B) Cousin
(C) Sister (D) Mother

Directions [Q. Nos. 83 to 85] are based on the following:

- Eleven students, A, B, C, D, E, F, G, H, I, J and K are sitting in the first row of the class facing the teacher.
- D who is the immediate left of F is second to the right of C.
- A is second to the right of E, who is at one of the ends.
- J is the immediate neighbor of A and B and third to the left of G.
- H is the immediate left of D and third to the right of I.

83. Who is sitting in the middle of the row?

(A) B (B) C
(C) G (D) I

84. If E and D, C and B, A and H, K and F interchange their position. Which of the following pairs of students are sitting at the ends?

(A) D and E (B) E and F
(C) D and K (D) K and F

85. Which of the following groups of friends is sitting to the right of G?

(A) CHDE (B) CHDF
(C) IBJA (D) None of the above

Directions [Q. Nos. 86 to 89] are based on the following:

- A group of six friends are sitting around a hexagonal table, each one at one corner of the hexagon.
- Ram is sitting opposite to Ramesh.
- Jyoti is sitting next to Seema.
- Neeta is sitting opposite to Seema but not next to Ram.
- Amrit has a person sitting between Ramesh and himself.

86. If Seema and Jyoti mutually interchange their positions, then who will be sitting opposite to Neeta?

(A) Jyoti (B) Ram
(C) Seema (D) Ramesh

87. If Neeta sits to the right of Amrit, then who is sitting to the left of Amrit?

(A) Ramesh (B) Neeta
(C) Jyoti (D) Ram

88. Who is sitting between Amrit and Ramesh?
(A) Neeta (B) Jyoti
(C) Seema (D) Ram

89. Who is sitting opposite to Jyoti?
(A) Ramesh (B) Neeta
(C) Amrit (D) Seema

90. A Group of 630 children are seated in n rows for a group photo session. Each row contains three less children than the row in front of it. Which one of the following number of rows is not possible?
(A) 3 (B) 4
(C) 5 (D) 6

English:

Directions (Q. Nos. 91 to 94) are based on the following. White cement is the basic raw material for producing cement tiles and cement paint which are used extensively in building construction. The main consumers of white cement are therefore, cement title and cement paint manufacturing units. These consumers, mostly in the small-scale sector, are today facing a major crisis because of a significant increase in the price of white cement during a short period The present annual licensed production capacity of white and grey cement in a country is approximately 3.5 lakh tones. The average demand is 2-2.5 lakh tones. This means that there is idle capacity to the tone of one lakh tones or more, the price is, therefore, not a phenomenon arising out of inadequate production capacity but evidently because of artificial scarcity created by the manufactures in their self-interest.

The main reason for the continuing spurt in cement price is its decontrol. As it is, there is stiff competition in the cement paint and tile manufacturing business. Any further price revision at this stage is bound to have a severe adverse impact on the market conditions. The Government should take adequate steps to ensure that suitable controls are brought in. Else it should allow import of cement.

91. Which of the following words has the opposite meaning as the word "basic" as used in the passage?
(A) Vital (B) Unimportant
(C) Acidic (D) Last

92. Why is the price of cement going up?
(A) Because the Government is controlling quota
(B) Because of export of white cement
(C) Because of the large usage of white cement
(D) None of these

93. What is the crisis faced by the cement tile manufactures as described in the passage?
(A) White cement prices are very high
(B) White cement is not of good quality
(C) White cement usage is high
(D) White cement is priced very low

94. Which of the following has the same meaning as the word 'artificial' as used in the passage?
(A) Deliberate (B) Prolonged
(C) Practical (D) Unnatural

95. Choose the words that accurately signifies a person who makes money by starting or running business
(A) Antreprenour (B) Andrapreneur
(C) Entraprenour (D) Entrepreneur

96. Which of the following is correct phrase to describe a group of insects?
(A) A flock of insects (B) A swarm of insects
(C) A school of insects (D) A shoal of insects

97. Which of the following has closet meaning to the word 'REPUTATION'?
(A) Character (B) Respect
(C) Fame (D) Honour

98. Which of the following words means 'Theatrical'?
(A) Thrilling (B) Histrionic
(C) Delicate (D) Delicious

99. Identify the word which is different from the rest of the words

(A) Indisputable (B) Uncertain
(C) Dubious (D) Doubtful

100. Select the pair that best expresses a relationship similar to the expressed in SCALE : TONE
(A) Physician : Medicine (B) Wave : Amplitude
(C) Spectrum : Colour (D) Rainbow : Shower

101. Choose the answer which best express the meaning the idom/phrase "to burn a hole in the pocket".
(A) Steal from someone's pocket
(B) To destroy other's belongings
(C) To be very miserly
(D) Money that is spent quickly

102. Choose the correct alternative to fill the blank.
My window look ……………… the garden.
(A) upon (B) out on
(C) in (D) at

103. Fill in the blanks with suitable article.
……………… darkest cloud has a silver lining
(A) An (B) A
(C) The (D) From

104. Fill in the blanks with appropriate adjective.
The steak is completely ………………, it is cold and tough.
(A) edible (B) erratic
(C) unswerving (D) thedible

105. Fill in the blank with a suitable preposition.
We have been looking for a new flat ………. ages.
(A) since (B) for
(C) during (D) in

106. Fill in the blanks with appropriate verb
Where is he? He should ………….. home hours ago.
(A) be (B) have been
(C) had been (D) were

107. Fill in the blank with appropriate question tag.
You shouldn't be here on a holiday, ………………
(A) shouldn't you (B) should you not
(C) wouldn't you (D) should you

108. Change the following sentence into passive sentence. They studied Mathematics last year.
(A) Mathematics was studied by them last year
(B) Mathematics were studied by them last year
(C) Mathematics has been studied by them last year
(D) Mathematics studied by them last year.

109. The meaning of the word "EGRESS" is
(A) Entrance (B) Exit
(C) Double (D) Program

110. Choose the answer which best expresses the meaning of the Idiom/phrase "Elbow room"
(A) Opportunity for freedom of action
(B) Special room for the guest
(C) To give enough space to move or work in
(D) To add a new room to the house

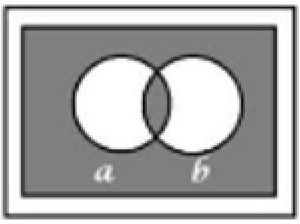
Computer:

111. Consider a hard disk with 16 recording surfaces (0-15) having 16384 cylinder (0-16383) and each cylinder contains 64 sectors (0-63). Data storage capacity in each sector in 512 bytes. Data are organised cylinder-wise and the addressing format is < cylinder no., surface no., sector no. >. A file of size 42797 KB is stored in the disk and the starting disk location of the file is < 1200, 9, 40>. What is the cylinder number of the last sector of the file, if it is stored in a continuous manner?

(A) 1284 (B) 1282
(C) 1286 (D) 1288

112. Consider the following minterm expression for F.
$F(P,Q,R,S) = \sum 0, 2, 5, 7, 8, 10, 13, 15$
The minterms 2, 7, 8 and 13 are 'do not care' terms.
The minimal sum-of-product form for F is
(A) $Q\bar{S} + \bar{Q}S$
(B) $QS + \bar{Q}\bar{S}$
(C) $\bar{Q}\bar{R}\bar{S} + \bar{Q}R\bar{S} + Q\bar{R}S + QRS$
(D) $\bar{P}\bar{Q}\bar{S} + \bar{P}QS + PQS + P\bar{Q}\bar{S}$

113. The Boolean expression represented by the following Venn diagram is



(A) a XOR b
(B) $a'b + ab'$
(C) $ab + a'b'$
(D) $(a + b')(a' + b)$

114. The range of n-bit signed magnitude representation is
(A) 0 to $2^n - 1$
(B) $-(2^{n-1} - 1)$ to $(2^{n-1} - 1)$
(C) $-(2^n - 1)$ to $(2^{n-1} - 1)$
(D) 0 to $(2^{n-1} - 1)$

115. A hard disk has a rotational speed of 6000 rpm. Its average latency time is
(A) $5 \times 10^{-3}\ sec$
(B) 0.05 sec
(C) 1 sec
(D) 0.5 sec

116. The 2's complement representation of the number $(-100)_{10}$ in an 8-bit computer is
(A) 10011011
(B) 01100100
(C) 11100100
(D) 10011100

117. The number of terms in the product of sum canonical form of $|(x_1 + x_2)\ (x_3\ x_4)|$ is
(A) 7
(B) 8
(C) 9
(D) 10

118. Find the odd one out:
(A) HTTP
(B) FCFS
(C) HTML
(D) TCP/IP

119. Consider the equation $(43)_x = (y3)_8$ where x and y are unknown. The number of possible solutions is
(A) 4
(B) 6
(C) 5
(D) 7

120. Subtract $(1010)_2$ from $(1101)_2$ using first complement
(A) $(1100)_2$
(B) $(0101)_2$
(C) $(1001)_2$
(D) $(0011)_2$

Answer Key

1. C	**13.** D	**25.** C	**37.** B	**49.** D	**61.** C	**73.** B	**85.** B	**97.** C	**109.** B
2. A	**14.** B	**26.** B	**38.** C	**50.** B	**62.** C	**74.** B	**86.** A	**98.** B	**110.** A, C
3. *	**15.** A	**27.** D	**39.** D	**51.** C	**63.** D	**75.** D	**87.** D	**99.** A	**111.** A
4. A	**16.** D	**28.** D	**40.** D	**52.** D	**64.** D	**76.** B	**88.** A	**100.** C	**112.** B
5. A	**17.** B	**29.** C	**41.** A	**53.** A	**65.** A	**77.** B	**89.** C	**101.** D	**113.** C
6. A	**18.** D	**30.** D	**42.** C	**54.** B	**66.** C	**78.** A	**90.** D	**102.** B	**114.** B
7. C	**19.** D	**31.** C	**43.** C	**55.** C	**67.** B	**79.** A	**91.** B	**103.** C	**115.** A
8. A	**20.** D	**32.** A	**44.** C	**56.** D	**68.** C	**80.** D	**92.** D	**104.** A	**116.** D
9. B	**21.** D	**33.** C	**45.** B	**57.** A	**69.** A	**81.** A	**93.** A	**105.** B	**117.** *
10. C	**22.** D	**34.** *	**46.** B	**58.** B	**70.** A	**82.** A	**94.** A	**106.** B	**118.** B
11. *	**23.** C	**35.** C	**47.** A	**59.** C	**71.** C	**83.** D	**95.** D	**107.** D	**119.** C
12. B	**24.** B	**36.** D	**48.** A	**60.** D	**72.** B	**84.** C	**96.** B	**108.** A	**120.** D

Solution

Mathematics

1. (c) Total number of peoples = 9

Number of ways in which 5 people groups can be selected from 9 people.

According to the question,

5 people are selected from 9 – 2 particulars people already in the same group

$\Rightarrow \quad {}^9C_5 - {}^7C_3 \qquad \Rightarrow \quad \dfrac{9!}{5!\,4!} - \dfrac{7!}{3!\,4!}$

$\Rightarrow \quad \dfrac{9 \times 8 \times 7 \times 6}{4 \times 3 \times 2} - \dfrac{7 \times 6 \times 5}{3 \times 2} \quad \Rightarrow \quad 126 - 35 \quad \Rightarrow \quad 91$

2. (a) By options,

here (b), (c) and (d) violates the conditions of logarithm because logarithm base become 1.

Hence, the correct answer is $\{2^{-\sqrt{2}}, 2^{\sqrt{2}}\}$.

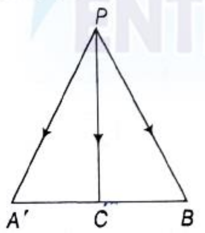
3. (*) Let the side of regular polygon is a.

Length of the each side of polygon is

$$a = 2R\sin\left(\frac{\pi}{n}\right)$$
$$= 2 \times 3\sin\left(\frac{180}{12}\right)$$
$$= 6\sin(15^\circ)$$
$$= 6\,\frac{(\sqrt{3}-1)}{2\sqrt{2}}$$
$$= \frac{3(\sqrt{3}-1)}{\sqrt{2}}$$
$$= \frac{3\sqrt{2}}{2}(\sqrt{3}-1)$$
$$= \frac{3\sqrt{6}-3\sqrt{2}}{2}$$

None option is correct.

4. (a) Given, C is the mid-point of AB.



Applying triangle law of addition of vector. In ΔPAC and PBC, we get

$$\mathbf{PA} + \mathbf{AC} = \mathbf{PC}$$

and $\qquad \mathbf{PB} + \mathbf{BC} = \mathbf{PC}$

On adding both equations,

$$\mathbf{PA} + \mathbf{AC} + \mathbf{PB} + \mathbf{BC} = \mathbf{PC} + \mathbf{PC}$$

$\Rightarrow \qquad \mathbf{PA} + \mathbf{PB} + \mathbf{AC} + \mathbf{BC} = 2\mathbf{PC}$

$\Rightarrow \qquad \mathbf{PA} + \mathbf{PB} + \mathbf{AC} - \mathbf{AC} = 2\mathbf{PC}$

($\because C$ is the mid-point of **AB**)

$\Rightarrow \qquad \mathbf{PA} + \mathbf{PB} = 2\mathbf{PC}$

5. (a) Given,

$$\mathbf{x}_1 = 52,\ \mathbf{x}_2 = 42 \text{ and } \mathbf{x} = 50$$

Let n_1 and n_2 be the number of boys and girls respectively in the class.

$\because \qquad \mathbf{x} = \dfrac{n_1\mathbf{x}_1 + n_2\mathbf{x}_2}{n_1 + n_2}$

$\Rightarrow \qquad 50 = \dfrac{52(n_1) + 42n_2}{n_1 + n_2}$

$\Rightarrow \qquad 50(n_1 + n_2) = 52n_1 + 42n_2$

$\Rightarrow \qquad 50n_1 + 50n_2 = 52n_1 + 42n_2$

$\Rightarrow \qquad 8n_2 = 2n_1$

$\Rightarrow \qquad n_1 = 4n_2$

So, percentage of boys $= \dfrac{n_1}{n_1 + n_2} \times 100$

$$= \frac{4n_2}{4n_2 + n_2} \times 100$$
$$= \frac{4n_2}{5n_2} \times 100$$
$$= \frac{4}{5} \times 100 = 80\%$$

6. (a) Total number of outcomes = selection of 4 caps from (2 blue + 4 red + 5 green + 1 yellow) 12 caps.

$$= {}^{12}C_4$$

Favourable number of outcomes $= {}^7C_4$

$\therefore$ Required probability $= \dfrac{{}^7C_4}{{}^{12}C_4}$

$$= \frac{\dfrac{7!}{4!\ 3!}}{\dfrac{12!}{4!\ 8!}} = \frac{7!}{3!} \times \frac{8!}{12!}$$

$$= \frac{8 \times 7 \times 6 \times 5 \times 4}{12 \times 11 \times 10 \times 9 \times 8} = \frac{7}{99}$$

7. (c) Given equation of line is $3x + 5y = k$ $\qquad$ ...(i)

comparing Eq. (i) with $lx + my + n = 0$, we get

$$l = 3,\ m = 5 \text{ and } n = -k$$

Equation of ellipse is

$$16x^2 + 25y^2 = 400 \qquad \text{...(ii)}$$

On divide Eq. (ii) by 400, we get

$$\frac{16x^2}{400} + \frac{25y^2}{400} = \frac{400}{400}$$

$$\Rightarrow \quad \frac{x^2}{25}+\frac{y^2}{16}=1$$

By comparing with standard form of ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$,

we get $\quad a=5, b=4$

$\because$ Condition of tangency

$$n^2=a^2l^2+b^2m^2$$
$$(-k)^2=(5)^2(3)^2+(4)^2(5)^2$$
$$\Rightarrow \quad k^2=225+400$$
$$\Rightarrow \quad k^2=625$$
$$\Rightarrow \quad K=\pm 25$$

8. (a) Given, $\quad X=\{4^n-3n-1, n\in N\}$ and

$$Y=\{9n-9, n\in N\}$$

$\because \quad 4^n-3n-1=(1+3)^n-3n-1$

$=({}^nC_0+{}^nC_1\times 3+{}^nC_2\times 3^2+\ldots+{}^nC_n3^n)-(3n+1)$

$[\because (1+x)^n=3^2({}^nC_0+{}^nC_3+\ldots {}^nC_n3^{n-2})]$

$\therefore X$ contains same multiples of 9.

Here it is clear that Y contains all multiples of 9 including 0.

$\therefore \quad X\cup Y$ equal to Y.

9. (b) Given, $\quad I=\int\left\{\frac{(\log x-1)}{1+(\log x)^2}\right\}^2 dx$

Let $\log x=t$

$$\Rightarrow \quad x=e^t$$
$$\Rightarrow \quad dx=e^t\,dt$$
$$\Rightarrow \quad I=\int e^t\frac{(t-1)^2}{(t^2+1)^2}dt$$
$$\Rightarrow \quad I=\int e^t\frac{t^2+1-2t}{(t^2+1)^2}dt$$
$$\Rightarrow \quad I=\int e^t\left[\frac{1}{t^2+1}+\frac{(-2t)}{(t^2+1)^2}\right]dt$$
$$\Rightarrow \quad I=\frac{e^t}{t^2+1}+c$$

$[\because \int e^x[f(x)+f'(x)]dx=e^xf(x)+C]$

$$\Rightarrow \quad I=\frac{x}{(\log x)^2+1}+C$$

10. (c) Given, $\quad U=\hat{\mathbf{i}}+2\hat{\mathbf{j}}-\hat{\mathbf{k}}$

$$V=-2\hat{\mathbf{i}}+3\hat{\mathbf{k}}$$

and $\quad w=7\hat{\mathbf{j}}-4\hat{\mathbf{k}}$

$\because$ Volume of parallelopiped $=U\cdot(V\times W)$

$$=\begin{vmatrix}1 & 2 & -1\\ -2 & 0 & 3\\ 0 & 7 & -4\end{vmatrix}$$

$=|1(-21)-2(8-0)-1(-14)|$

$=|-21-16+14|$

$=|14-37|$

$=|-23|$

$=23$ $\quad$ [since volume cannot be negative]

11. (*) Let the points are $A(1,-1,0)$, $B(-2,1,-1)$ and $C(-1,1,2)$

$\therefore$ **AB** $=(-3,2,-1)$ and **AC** $=(-2,2,2)$

The vector perpendicular to plane is

$$=\mathbf{AB}\times\mathbf{AC}$$

So, $\mathbf{AB}\times\mathbf{AC}=\begin{vmatrix}\hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}}\\ -3 & 2 & -1\\ -2 & 2 & 2\end{vmatrix}$

$$=\hat{\mathbf{i}}(4+2)-\hat{\mathbf{j}}(-6-2)+\hat{\mathbf{k}}(-6+4)]$$
$$=6\hat{\mathbf{i}}+8\hat{\mathbf{j}}-2\hat{\mathbf{k}}$$

None option is correct.

12. (b) Given, equation of circles are,

$$2x-3y+12=0 \quad \ldots(i)$$
$$x+4y-5=0 \quad \ldots(ii)$$

On solving Eq. (i) and Eq. (ii), we get

$$x=-3 \text{ and } y=2$$

Hence, coordinate of centre is $(-3,2)$.

Given, area of circle $=154$ sq units

$\Rightarrow \quad \pi r^2=154$ [where r is radius of circle]

$$\Rightarrow \quad \frac{22}{7}\times r^2=154$$
$$\Rightarrow \quad r^2=49$$
$$\Rightarrow \quad r=7$$

Here, centre is $(-3,2)$ and radius $=7$

Then, required of equation of circle is

$$\{x-(-3)\}^2+(y-2)^2=(7)^2$$
$$\Rightarrow \quad (x+3)^2+(y-2)^2=49$$
$$\Rightarrow \quad x^2+9+6x+y^2+4-4y=49$$
$$\Rightarrow \quad x^2+y^2+6x-4y+13-49=0$$
$$\Rightarrow \quad x^2+y^2+6x-4y-36=0$$

13. (d) Here, $\quad I=\int\frac{x^2-1}{x^3\sqrt{2x^4-2x^2+1}}dx$

[On dividing numerators and denominators by x^5]

$$\Rightarrow \quad I=\int\frac{\frac{x^2-1}{x^5}}{\frac{x^3\sqrt{2x^4-2x^2+1}}{x^5}}dx$$

$$=\int\frac{\frac{1}{x^3}-\frac{1}{x^5}}{\frac{x^2\sqrt{2-\frac{2}{x^2}+\frac{1}{x^4}}}{x^2}}dx$$

$$=\int\frac{\frac{1}{x^3}-\frac{1}{x^5}}{\sqrt{2-\frac{2}{x^2}+\frac{1}{x^4}}}dx$$

Let $\left(2-\frac{2}{x^2}+\frac{1}{x^4}\right)=t$

$$\Rightarrow \left(\frac{4}{x^3}-\frac{4}{x^5}\right)dx=dt$$

$$\therefore \quad I=\frac{1}{4}\int\frac{dt}{\sqrt{C}}$$
$$=\frac{1}{4}\int t^{-1/2}\,dt$$
$$=\frac{1}{4}\frac{t^{1/2}}{\frac{1}{2}}+C \quad \text{[Here } C \text{ is integration constant]}$$
$$=\frac{1}{4}\times 2t^{1/2}+C$$
$$=\frac{1}{2}t^{1/2}+C$$

On substituting the value of t, then

$$I=\frac{1}{2}\sqrt{2-\frac{2}{x^2}+\frac{1}{x^4}}+C$$
$$=\frac{1}{2}\sqrt{\frac{2x^4-2x^2+1}{x^4}}+C$$
$$=\frac{1}{2}\frac{\sqrt{2x^4-2x^2+1}}{x^2}+C$$
$$=\frac{\sqrt{2x^4-2x^2+1}}{2x^2}+C$$

14. (b) Given, $|\mathbf{a}|=|\mathbf{b}|=|\mathbf{a}+\mathbf{b}|=\alpha$

$$\because \quad |\mathbf{a}+\mathbf{b}|^2=|\mathbf{a}|^2+|\mathbf{b}|^2+2\,\mathbf{a}\cdot\mathbf{b}$$
$$\Rightarrow \quad \alpha^2=\alpha^2+\alpha^2+2\mathbf{a}\cdot\mathbf{b}$$
$$\Rightarrow \quad \mathbf{a}\cdot\mathbf{b}=-\frac{\alpha^2}{2} \quad \ldots(\text{i})$$

Then, $|\mathbf{a}-\mathbf{b}|^2=|\mathbf{a}|^2+|\mathbf{b}|^2-2\,\mathbf{a}\cdot\mathbf{b}$

$$=\alpha^2+\alpha^2-2\cdot\frac{(-\alpha^2)}{2} \quad \text{[From Eq. (i)]}$$
$$=\alpha^2+\alpha^2+\alpha^2$$
$$|\mathbf{a}-\mathbf{b}|^2=3\alpha^2$$
$$\Rightarrow \quad |\mathbf{a}-\mathbf{b}|=\sqrt{3\alpha^2}$$
$$=\sqrt{3}\alpha$$

15. (a)

16. (d) Given equation of pair of straight line is

$$2x^2+7xy+3y^2+8x+14y+\lambda=0 \quad \ldots(\text{i})$$

On comparing Eq. (i) by
$ax^2+2hxy+by^2+2gx+2fy+c=0$, we get

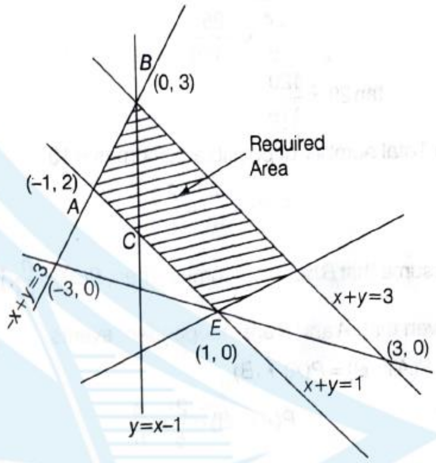
$$a=2,\ b=3,\ c=\lambda,\ h=\frac{7}{2},\ g=4 \text{ and } f=7$$

Eq. (i) will represents a pair of straight line, if

$$abc+2fgh-af^2-bg^2-ch^2=0$$
$$\Rightarrow 2\times3\times\lambda+2\times7\times4\times\frac{7}{2}-2\times7^2-3\times4^2-\lambda\times\left(\frac{7}{2}\right)^2=0$$
$$\Rightarrow \quad 6\lambda+196-98-48-\lambda\times\frac{49}{4}=0$$
$$\Rightarrow \quad \frac{24\lambda-49\lambda}{4}+50=0$$
$$\Rightarrow \quad \frac{-25\lambda}{4}=-50$$
$$\Rightarrow \quad -25\lambda=-200$$
$$\Rightarrow \quad \lambda=8$$

17. (b) $y=|x-1|=\begin{cases}x-1: x\ge1\\1-x: x<1\end{cases} \quad \ldots(\text{i})$

and $y=3-|x|=\begin{cases}3-x: x\ge0\\3-x: x<0\end{cases} \quad \ldots(\text{ii})$



Clearly, the two curves meet at $(-1, 2)$ and $(2, 1)$.

Required area

$$=\int_{-1}^{0}(y_2-y_1)\,dx+\int_0^1(y^2-y_1)\,dx+\int_1^2(y_2-y_1)\,dx$$
$$=\int_{-1}^{\infty}[3+x-(1-x)]\,dx+\int_0^1[(3-x)-(1-x)]\,dx$$
$$+\int_1^2[(3-x)(x-1)\,dx$$
$$=4$$

18. (d) Given that, $a=4, b=3$

and $\angle BAC=\angle A=60^\circ$

$$\because \quad \cos A=\frac{b^2+c^2-a^2}{2bc}$$
$$\Rightarrow \quad \cos 60^\circ=\frac{3^2+c^2-4^2}{2\times3\times c}$$
$$\Rightarrow \quad \frac{1}{2}=\frac{9+c^2-16}{6c}$$
$$\Rightarrow \quad c^2-7=3c$$
$$\Rightarrow \quad c^2-3c-7=0$$

19. (d) We have, $\cos\theta=\frac{5}{13}$ $\left[\text{where, } \frac{3\pi}{2}<\theta<2\pi\right]$

By trigonometric ratios,

$$\tan\theta=-\frac{12}{5} \quad [\because \theta \text{ belongs to fourth quadrant}]$$
$$\because \quad \tan 2\theta=\frac{2\tan\theta}{1-\tan^2\theta}$$

$\therefore \quad \tan 2\theta = \dfrac{2 \times \left(\dfrac{-12}{5}\right)}{1 - \left(\dfrac{-12}{5}\right)^2}$

$= \dfrac{-\dfrac{24}{5}}{1 - \dfrac{144}{25}}$

$= \dfrac{-\dfrac{24}{5}}{-\dfrac{119}{25}}$

$= -\dfrac{24}{5} \times \dfrac{25}{-119}$

$\Rightarrow \quad \tan 2\theta = \dfrac{120}{119}$

20. (d) Total number of possible outcomes = 10

$\therefore \quad P(A) = \dfrac{4}{10} = \dfrac{2}{5}$

Assume that B has n outcomes. Then, $P(B) = \dfrac{n}{10}, [0 < n < 10]$

Given that, A and B are independent events.

$\therefore \quad P(A \cap B) = P(A) \cdot P(B)$

$\Rightarrow \quad P(A \cap B) = \dfrac{2}{5} \times \dfrac{n}{10}$

$\Rightarrow \quad P(A \cap B) = \dfrac{\dfrac{2n}{5}}{10}$

$\because \dfrac{2n}{5}$ is an integer which is lies between 0 and 10 and $\dfrac{n}{5}$ is an integers which is lies between 0 and 5.

$\therefore \quad n = 5, 10$

21. (d) Given that **a**, **b** and **c** are non-zero vectors and **a** + 2**b** is collinear with **c** and **b** + 3**c** is collinear with **a**.

$\therefore \quad \mathbf{a} + 2\mathbf{b} = \lambda\mathbf{c} \quad \ldots\text{(i)}$

and $\quad \mathbf{b} + 3\mathbf{c} = \mu\mathbf{a} \quad \ldots\text{(ii)}$

[for same λ, μ]

Put the value of **b** in Eq. (i) by Eq (ii) , we get

$\mathbf{a} + 2(\mu\mathbf{a} - 3\mathbf{c}) = \lambda\mathbf{c}$

$\Rightarrow \quad \mathbf{a} + 2\mu\mathbf{a} - 6\mathbf{c} = \lambda\mathbf{c}$

$\Rightarrow \quad \mathbf{a}(1 + 2\mu) - \mathbf{c}(\lambda + 6) = 0$

[$\because$ **a** and **c** are non-collinear]

$\Rightarrow \quad 1 + 2\mu = 0$ and $\lambda + 6 = 0$

$\Rightarrow \quad \lambda = -6$

and $\quad \mu = -\dfrac{1}{2}$

$\because \quad \mathbf{a} + 2\mathbf{b} = \lambda\mathbf{c}$

$\therefore \quad \mathbf{a} + 2\mathbf{b} = -6\mathbf{c}$

$\Rightarrow \quad \mathbf{a} + 2\mathbf{b} + 6\mathbf{c} = 0$

22. (d) Given equation is

$x^2 - (a-2)x - (a+1) = 0$

Here, $\quad a = 1, b = -(a-2)$

and $\quad c = -(a+1)$.

Let the roots of equation is α and β.

$\alpha + \beta = -\dfrac{b}{a}$

$= a - 2$

and $\quad \alpha\beta = \dfrac{c}{a}$

$= -(a+1)$

$\Rightarrow \quad \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$= (a-2)^2 - 2\{-(a+1)\}$

$\Rightarrow \quad \alpha^2 + \beta^2 = (a-2)^2 + 2(a+1)$

$= a^2 + 4 - 4a + 2a + 2$

$= a^2 - 2a + 1 + 5$

$\Rightarrow \quad \alpha^2 + \beta^2 = (a-1)^2 + 5$

Clearly, $\quad \alpha^2 + \beta^2 \geq 5$

So, the minimum value of $\alpha^2 + \beta^2$ is 5 which it atains at $a = 1$.

23. (c) Given that,

$P(A \cup B) = 0.6 = \dfrac{3}{5}$

$P(A \cap B) = 0.3 = \dfrac{3}{10}$

Then, by probability law,

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$\Rightarrow \quad \dfrac{3}{5} = P(A) + P(B) - \dfrac{3}{10}$

$\Rightarrow \quad P(A) + P(B) = \dfrac{3}{5} + \dfrac{3}{10}$

$= \dfrac{30 + 15}{50} = \dfrac{45}{50}$

$\Rightarrow \quad P(A) + P(B) = \dfrac{9}{10}$

$\Rightarrow \quad 1 - P(\bar{A}) + 1 - P(\bar{B}) = \dfrac{9}{10} \quad [\because P(A) + P(\bar{A}) = 1]$

$\Rightarrow \quad 2 - P(\bar{A}) - P(\bar{B}) = \dfrac{9}{10}$

$\Rightarrow \quad P(\bar{A}) + P(\bar{B}) = 2 - \dfrac{9}{10}$

$= \dfrac{20 - 9}{10}$

$\Rightarrow \quad P(\bar{A}) + P(\bar{B}) = \dfrac{11}{10} = 1.1$

24. (b) Let the total number of subset in set $A = 2^m$ and subset in set $B = 2^n$

According to the given condition,

$$2^m = 2^n + 56$$

$$\Rightarrow \quad 2^m - 2^n = 56$$

$$\Rightarrow \quad 2^n(2^{m-n} - 1) = 8 \times 7$$

$$\Rightarrow 2^n(2^{m-n} - 1) = 2^3 \times (2^3 - 1)$$

On comparing both sides, we get

$$n = 3 \text{ and } m - n = 3$$

$$\Rightarrow \quad m - 3 = 3$$

$$\Rightarrow \quad m = 6$$

So, $m = 6$ and $n = 3$

25. (c) Given that,

$$P(\text{Speak truth}) = P(A) = \frac{4}{5}$$

and $\quad P(B) = \frac{3}{4}$

$\therefore \quad P(\bar{A}) = \frac{1}{5}$

and $\quad P(\bar{B}) = \frac{1}{4} \qquad [\because P(A) + P(\bar{A}) = 1]$

Then, the probability when they contradict each other

$$= P(A) \cdot P(\bar{B}) + P(B) \cdot P(\bar{A})$$

$$= \frac{4}{5} \times \frac{1}{4} + \frac{3}{4} \times \frac{1}{5}$$

$$= \frac{4}{20} + \frac{3}{20} = \frac{7}{20}$$

26. (b) Let $S = \frac{1}{\sqrt{1} + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \frac{1}{\sqrt{3} + \sqrt{4}} + \ldots + \frac{1}{\sqrt{80} + \sqrt{81}}$

Applying rationalisation rule in each term, we get

$$S = \frac{\sqrt{2} - \sqrt{1}}{2 - 1} + \frac{\sqrt{3} - \sqrt{2}}{3 - 2} + \ldots + \frac{\sqrt{81} - \sqrt{80}}{81 - 80}$$

$$= \sqrt{2} - 1 + \sqrt{3} - \sqrt{2} + \ldots + \sqrt{81} - \sqrt{80}$$

$$= 9 - 1 = 8$$

27. (d) Given that, $f(x) = \begin{cases} x^2 - 1, & x < 3 \\ 2ax, & x \geq 3 \end{cases}$

Since, $f(x)$ is continuous at $x = 3$, then

$$f(3) = \lim_{h \to 0} f(3 + h) = \lim_{h \to 0} f(3 - h)$$

$$\Rightarrow \lim_{h \to 0} 2a(3 + h) = \lim_{h \to 0} [(3 - h)^2 - 1]$$

$$\Rightarrow \lim_{h \to 0} (6a + 2ah) = \lim_{h \to 0} [9 + h^2 - 6h - 1]$$

$$\Rightarrow \lim_{h \to 0} (6a + 2ah) = \lim_{h \to 0} [8 + h^2 - 6h]$$

$$\Rightarrow \quad 6a = 8$$

$$\Rightarrow \quad a = \frac{8}{6} = \frac{4}{3}$$

28. (d) Since, each person can apply for any house.

Therefore, total number of ways in which three person can apply for three houses

$$= 3 \times 3 \times 3$$

$$= 27 \text{ ways}$$

As given that three person can apply for house 1 or house 2 or house 3 among three house.

So, three person can apply for the same house in 3 ways.

$\therefore$ Favourable number of ways = 3

Hence, required probability

$$= \frac{\text{Favourable number of ways}}{\text{Total number of ways}}$$

$$= \frac{3}{27} = \frac{1}{9}$$

29. (c) Required probability

$= 1 - P$(none of the selected houses is a winning house)

$$= 1 - \left(\frac{4}{5} \times \frac{3}{4}\right)$$

$$= 1 - \frac{3}{5} = \frac{2}{5}$$

30. (d) Given,

$$3^x = 4^{x-1}$$

On taking log both sides, we get

$$\log(3^x) = \log 4^{(x-1)}$$

$$\Rightarrow \quad x \log 3 = (x - 1) \log 4 \quad [\because \log m^n = n \log m]$$

$$\Rightarrow \quad x \log 3 = x \log 4 - \log 4$$

$$\Rightarrow \quad x \log 3 - \log 4 = -\log 4$$

$$\Rightarrow \quad x(\log 3 - \log 4) = -\log 4$$

$$\Rightarrow \quad -x(\log 4 - \log 3) = -\log 4$$

$$\Rightarrow \quad x = \frac{\log 4}{\log 4 - \log 3}$$

$$= \frac{\log 2^2}{\log 2^2 - \log 3}$$

$$= \frac{2 \log 2}{2 \log 2 - \log 3}$$

$$= \frac{\frac{2 \log 2}{\log 3}}{\frac{2 \log 2 - \log 3}{\log 3}}$$

[On dividing numerator and denominator by log 3]

$$\Rightarrow \quad x = \frac{2 \log_3 2}{2 \log_3 2 - 1}$$

31. (c) We have, $[A]_{x \times (x+5)}$

[where x = number at rows and $x + 5$ = number of column]

and $[B]_{y \times (11-y)}$

[where y = number of rows and $11 - y$ = number of columns]

According to the matric rule AB exist if

$\Rightarrow$ Number of column of A = number of raw of B

$$\Rightarrow \quad x + 5 = y \quad \ldots(i)$$

and BA exist if

$\Rightarrow$ Number of column of B = Number of raw of A

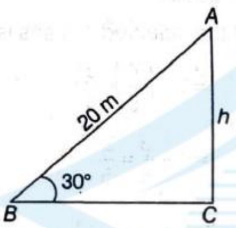
$$\Rightarrow \quad 11 - y = x \quad \ldots(ii)$$

By solving Eq. (i) and Eq. (ii), we get

$$x = 3 \text{ and } y = 8$$

32. (a) Let the length of rope $(AB) = 20$ m

and length of vertical pole = hm



In ΔABC, by sine rule, we get

$$\sin 30^\circ = \frac{AC}{AB}$$

$$\Rightarrow \quad \frac{1}{2} = \frac{h}{20}$$

$$\Rightarrow \quad 2h = 20$$

$$\Rightarrow \quad h = \frac{20}{2}$$

$$= 10 \text{ m}$$

33. (c) There are given n equally spaced points $1, 2, \ldots\ldots, n$

Given 15 is directly opposite to the point 49.

$\therefore$ other point will be $49 - 15 = 34$

But two points must be exclude.

So, there are 32 points which apposite

Then, total number of points are

$$= 32 \times 2 + 2$$

$$= 64 + 2 = 66$$

34. (*) Given, $S = \{1, 2, \ldots, n\}$

First chose a k element subset B of S. This can be done in $\binom{n}{k}$ ways.

Given the subset B, A has $2^K - 1$ options (since the numbers of proper subset of a k- element let is $2^k - 1$)

Hence, the total number of pairs of sets (A, B) such that $A \subseteq B$ is given by

$$\sum_{k=0}^{n} (2^K - 1)\binom{n}{k} = \sum_{k=0}^{n} \binom{n}{k} 2^k - \sum_{k=0}^{n} \binom{n}{k}$$

$$= 3^n - 2^n$$

None option is correct.

35. (c) Given equation is

$$4^x - 3(2^{x+3}) + 128 = 0 \quad \ldots(i)$$

Put $2^x = y$ in Eq. (i), we get

$$\Rightarrow \quad (2^x)^2 - 3(2^x \cdot 2^3) + 128 = 0$$

$$\Rightarrow \quad y^2 - 3(8y) + 128 = 0$$

$$\Rightarrow \quad y^2 - 24y + 128 = 0$$

$$\Rightarrow \quad y^2 - 16y - 8y + 128 = 0$$

$$\Rightarrow \quad y(y - 16) - 8(y - 16) = 0$$

$$\Rightarrow \quad (y - 16)(y - 8) = 0$$

$$\Rightarrow \quad y = 16, 8$$

$$\Rightarrow \quad 2^x = 16 \text{ or } 2^x = 8$$

$$\Rightarrow \quad 2^x = 2^4 \text{ or } 2^x = 2^3$$

$$\Rightarrow \quad x = 4, 3$$

$\therefore$ Sum of the root $= 4 + 3 = 7$

36. (d) Given equation of lines is

$$x^2 - 2cxy - 7y^2 = 0 \quad \ldots(i)$$

Let the m_1 and m_2 be the slopes of Eq. (i), then

$$m_1 + m_2 = -\frac{2c}{7}$$

and

$$m_1 m_2 = -\frac{1}{7}$$

According to given condition,

$$m_1 + m_2 = 4m_1 m_2$$

$$\Rightarrow \quad \frac{-2c}{7} = 4\left(-\frac{1}{7}\right)$$

$$\Rightarrow \quad -\frac{2c}{7} = -\frac{4}{7}$$

$$\Rightarrow \quad 2c = 4$$

$$\Rightarrow \quad c = 2$$

37. (b) Given system of equations is

$$x + y + 2z = a$$

$$x + z = b$$

$$2x + y + 3z = c$$

Applying matrix method, we get augmented matrix in the following form

$$= \begin{bmatrix} 1 & 1 & 2 & : & a \\ 1 & 0 & 1 & : & b \\ 2 & 1 & 3 & : & c \end{bmatrix}$$

$$=\begin{bmatrix}1 & 1 & 2 & : & a \\ 0 & 1 & 1 & : & a-b \\ 0 & 1 & 1 & : & 2a-c\end{bmatrix}$$

[applying $R_2 \to R_1 - R_2$ and $R_3 \to 2R_1 - R_3$]

$$=\begin{bmatrix}1 & 1 & 2 & : & a \\ 0 & 1 & 1 & : & a-b \\ 0 & 0 & 0 & : & a+b-c\end{bmatrix}$$

[applying $R_3 \to R_3 - R_2$]

If solution exist, then

$\Rightarrow \quad a+b-c=0$

$\Rightarrow \quad a+b=c$

38. (c) We have, $f(x)=x^2-bx+c$ and $b+c=35$

According to the condition $b+c=35$,

Let assume that 2 and 11 be the prime root of above equation, then $(x-2)(x-11)=0$

$\Rightarrow \quad x^2-2x-11x+22=0$

$\Rightarrow \quad x^2-13x+22=0$

Which satisfies the condition $b+c=35$.

On comparing both equations, we get

$b=13$ and $c=22$

Global minimum $f'(x)=0$

$\Rightarrow \quad 2x-13=0 \quad \Rightarrow \quad x=\frac{13}{2}$

$\therefore \quad f''\left(\frac{13}{2}\right)>0$

So, $f\left(\frac{13}{2}\right)=\left(\frac{13}{2}\right)^2-13\times\frac{13}{2}+22$

$=\frac{169}{4}-\frac{169}{2}+22=-\frac{81}{4}$

39. (d) Given focus is $(-1,1)$ and equation of directrix is $4x+3y-24=0$

The equation of axis which perpendicular to directrix is

$3x-4y+c=0$

Point $(-1,1)$ lies on above equation,

$3(-1)-4(1)+c=0$

$\Rightarrow \quad -3-4+c=0$

$\Rightarrow \quad c-7=0$

$\Rightarrow \quad c=7$

Then, equation of axis is $3x-4y+7=0$

Since, vertex lie on given axis.

Therefore, only option (d) satisfy the above equation.

Hence, vertex will be $\left(1,\frac{5}{2}\right)$.

40. (d) We have, $x^2-x\sin x-\cos x=0$

for any positive root in $(-\infty,\infty)$,

$f'(x)=2x-\sin x-x\cos x+\sin x=0$

$\Rightarrow \quad 2x-x\cos x=0$

$\Rightarrow \quad x(2-\cos x)=0$

$\cos x=2$ [which is not possible]

Hence, number of root will be zero.

41. (a) Books on fairy tales can be arranged in 4! ways novels can be arranged in 5! ways.

Books on plays can be arranged in 3! ways.

Number of arrangement in which fairy tales books are arranged first, then novels and after that books on plays

$=4!\times5!\times3!$

$=24\times120\times6=17280$

42. (c) Given that,

A has 100 observations from 101, 102, ... 200 and B has 100 observations from 151, 152, ..., 250.

Here, It is given that each of the two population has 100 observations which are 100 consecutive integers. So, sum of the squares of deviations from their respective mean are same.

So, $\quad V_A=V_B$

$\therefore \quad \frac{V_A}{V_B}=1$

43. (c) Given $|\mathbf{a}+\mathbf{b}|=\sqrt{29}$

and $\mathbf{a}\times(2\hat{i}+3\hat{j}+4\hat{k})=(2\hat{i}+3\hat{j}+4\hat{k})\times\mathbf{b}$

$\Rightarrow \quad \mathbf{a}\times(2\hat{i}+3\hat{j}+4\hat{k})=-\mathbf{b}\times(2\hat{i}+3\hat{j}+4\hat{k})$

$\Rightarrow \quad (\mathbf{a}+\mathbf{b})\times(2\hat{i}+3\hat{j}+4\hat{k})=0$

$\Rightarrow \quad \mathbf{a}+\mathbf{b}=\lambda(2\hat{i}+3\hat{j}+4\hat{k})$

$\Rightarrow \quad |\mathbf{a}+\mathbf{b}|=|\lambda|\sqrt{4+9+16}$

$\Rightarrow \quad \sqrt{29}=|\lambda|\sqrt{29}$

$\Rightarrow \quad |\lambda|=1$

$\Rightarrow \quad \lambda=\pm1$

$\therefore \quad \mathbf{a}+\mathbf{b}=\pm(2\hat{i}+3\hat{j}+4\hat{k})$

Hence, $\quad (\mathbf{a}+\mathbf{b})\cdot(-7\hat{i}+2\hat{j}+3\hat{k})$

$\Rightarrow \quad (2\hat{i}+3\hat{j}+4\hat{k})\cdot(-7\hat{i}+2\hat{j}+3\hat{k})$

$\Rightarrow \quad -14+6+12=4$

44. (c) Given, $\Sigma x_i^2=400$ and $\Sigma x_i=80$

Applying variance rule,

$$\frac{\Sigma x_i^2}{n}\geq\frac{(\Sigma x_i)^2}{n^2}$$

$\Rightarrow \quad \frac{400}{n}\geq\frac{80\times80}{n^2}$

$\Rightarrow \quad n\geq\frac{6400}{400}$

$\Rightarrow \quad n\geq16$

So, possible value of n can be 20.

45. (b) The vertices of any rectangle inscribed in an ellipse is given by

$$(\pm a\cos\theta, \pm b\sin\theta)$$

The area of the rectangle is given by

$$A(\theta) = 4ab\cos\theta\sin\theta$$
$$= 2ab\sin 2\theta$$

So, the area is maximum when $\sin(2\theta) = 1$

Hence, the maximum area is when $2\theta = \frac{\pi}{2}$.

i.e. $\theta = \frac{\pi}{4}$ the maximum area is $A = 2ab$

46. (b) Equation of tangent to parabola $y^2 = 8ax$ is $y = mx + \frac{2a}{m}$.

This tangent is also touches the circle $x^2 + y^2 = 2a^2$.

$\therefore$ Length of perpendicular = Radius

$$\Rightarrow \quad \frac{\frac{2a}{m}}{\sqrt{m^2+1}} = \sqrt{2}\,a$$
$$\Rightarrow \quad \frac{4a^2}{m^2(m^2+1)} = 2a^2$$
$$\Rightarrow \quad m^2(m^2+1) = 2$$
$$\Rightarrow \quad m^4 + m^2 - 2 = 0$$
$$\Rightarrow \quad (m^2-1)(m^2+2) = 0$$
$$\Rightarrow \quad m^2 = 1$$
$$\therefore \quad m = \pm 1$$

Hence, equation of common tangent is $y = \pm(x + 2a)$

47. (a) Given $a_1, a_2, \ldots, a_n$ are in AP and $a_1 = 0$

$\therefore \quad a_2 = d, \quad a_3 = 2d$

$$\because \left(\frac{a_3}{a_2} + \frac{a_4}{a_3} + \ldots + \frac{a_n}{a_{n-1}}\right) - a_2\left(\frac{1}{a_2} + \frac{1}{a_3} + \ldots + \frac{1}{a_{n-2}}\right)$$

$\therefore$

$$\left(\frac{2d}{d} + \frac{3d}{2d} + \frac{4}{3} + \ldots \frac{(n-1)d}{(n-1)d}\right) - d\left(\frac{1}{d} + \frac{1}{2d} + \ldots + \frac{1}{(n-3)d}\right)$$

$$= \left[1 + 1 + 1 + \frac{1}{2} + 1 + \frac{1}{3} + \ldots + 1 + \frac{1}{n-2}\right] - \left[1 + \frac{1}{2} + \frac{1}{3} + \ldots, + \frac{1}{n-3}\right]$$

$$= \left[1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{n-3} + \frac{1}{n-2} + n - 2\right] - \left[1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{n-3}\right]$$

$$= (n-2) + \frac{1}{(n-2)}$$

48. (a) Given,

$$\cos 20° + \cos 100° + \cos 140°$$
$$\Rightarrow \quad (\cos 20° + \cos 100°) + \cos 140°$$
$$= 2\cos\left(\frac{20° + 100°}{2}\right)\cos\left(\frac{20° - 100°}{2}\right) + \cos 140°$$
$$\left[\because \cos C + \cos D = 2\cos\frac{C+D}{2}\cos\frac{C-D}{2}\right]$$
$$= 2\cos 60°\cos(-40°) + \cos 140°$$
$$= 2\cos 60°\cos 40° + \cos 140° \qquad [\because \cos(-\theta) = \cos\theta]$$
$$= 2 \times \frac{1}{2}\cos 40° + \cos 140°$$
$$= 2\cos\left(\frac{140° + 40°}{2}\right)\cos\left(\frac{140° - 40°}{2}\right)$$
$$= 2\cos 90° \cdot \cos 50° \qquad [\because \cos 90° = 0]$$
$$\Rightarrow 0$$

49. (d) From the option, of all the given options (a), (b), (c) are different from given orders of word.

50. (b) Given equation of ellipse is

$$\frac{x^2}{16} + \frac{y^2}{b^2} = 1 \qquad \ldots\text{(i)}$$

and equation of hyperbola

$$\frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{25}$$
$$\Rightarrow \quad \frac{x^2}{\frac{144}{25}} - \frac{y^2}{\frac{81}{25}} = 1 \qquad \ldots\text{(ii)}$$

By comparing Eq. (ii) with $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, we get

$$a^2 = \frac{144}{25},$$
$$b^2 = \frac{81}{25}$$

So, foci of the ellipse are $(\pm\sqrt{16 - b^2}, 0)$ and hyperbola are

$$\left(\pm\sqrt{\left(\frac{12}{5}\right)^2 + \left(\frac{9}{5}\right)^2}, 0\right)$$

If they coincide, then

$$16 - b^2 = \frac{144}{25} + \frac{81}{25}$$
$$\Rightarrow \quad 16 - b^2 = \frac{225}{25}$$
$$\Rightarrow \quad 16 - b^2 = 9$$
$$\Rightarrow \quad b^2 = 16 - 9 = 7$$

Analytical Ability and Logical Reasoning

51. (c) Total number of fruits in the basket

$= 19 \times 1 - 4 \times 2$

$= 19 - 8 = 11$

52. (d) Total number of two's = total number of orange = 6

Total number of four = 4

$\therefore$ Total number of oranges in the basket

$= 6 - 4 = 2$

Solutions (Q. Nos. 53-56) From the given information the order will be as

8	K
7	N
6	J
5	Q
4	L
3	O
2	M
1	P

53. (a) After interchanging the places, O lives between P and M.

54. (b) Only two people live between M and Q.

55. (c) K lives on floor number 8.

56. (d) Only O and M are adjacent to each other.

Solutions (Q. Nos. 57-59)

Forwards - A, C, E and J

Point Guard - B, G and H

Defenders - D, F and I

57. (a) The largest possible team shall be of 6.

F, J, A, D, G and B/E.

58. (b) H cannot be included in 6 member team as if H is included then A and D will not be in the team.

59. (c) G will be a member of 6 member team.

60. (d) Through option we can check the possibilities. With option (d), we have

B — B — B — B

S — S — S

So, each boy has 3 sisters and 3 brothers and each girl has 2 sisters and 4 brothers.

61. (c)

13 $\boxed{4}$ – $\boxed{\text{good}}$ and tasty

$\boxed{4}$ 7 8 – see $\boxed{\text{good}}$ $\boxed{\text{picture}}$

7 29 – $\boxed{\text{picture}}$ are faint

As, good = 4

Picture = 7

So, see = 8

62. (c) As, $E \rightarrow 05 \rightarrow 0 + 5 = 5$

$X \rightarrow 24 \rightarrow 2 + 4 = 6$

$A \rightarrow 01 \rightarrow 0 + 1 = 1$

$M \rightarrow 13 \rightarrow 1 + 3 = 4$

$I \rightarrow 09 \rightarrow 0 + 9 = 9$

$N \rightarrow 14 \rightarrow 1 + 4 = 5$

$A \rightarrow 01 \rightarrow 0 + 1 = 1$

$T \rightarrow 20 \rightarrow 2 + 0 = 2$

$I \rightarrow 09 \rightarrow 0 + 9 = 9$

$O \rightarrow 15 \rightarrow 1 + 5 = 6$

$N \rightarrow 14 \rightarrow 1 + 4 = 5$

Similarly, $G \rightarrow = 07 \rightarrow 0 + 7 = 7$

$O \rightarrow 15 \rightarrow 1 + 5 = 6$

$V \rightarrow 22 \rightarrow 2 + 2 = 4$

$E \rightarrow 05 \rightarrow 0 + 5 = 5$

$R \rightarrow 18 \rightarrow 1 + 8 = 9$

$N \rightarrow 14 - \rightarrow 1 + 4 = 5$

$M \rightarrow 13 \rightarrow 1 + 3 = 4$

$E \rightarrow = 05 \rightarrow 0 + 5 = 5$

$N \rightarrow 14 \rightarrow 1 + 4 = 5$

$T \rightarrow 20 \rightarrow 2 + 0 = 2$

GOVERNMENT – 7645954552

63. (d) None of the given statements match the facts.

64. (d) There are 19 numbers which has 6 as a digit when unit digit 6 is replaced by 9, then sum of numbers will be increase by $= 3 \times 10 = 30$

From 60 to 69 there are 10 numbers which have ten's digit 6.

Here, the increase will be $30 \times 10 = 300$

$\therefore$ Total increase $= 30 + 300 = 330$

65. (a) Let the number of children be x.

$\therefore \quad x \times \frac{x}{5} = 405$

$\Rightarrow \quad x^2 = 5 \times 405 = 2025$

$\Rightarrow \quad x = \sqrt{2025} = 45$

66. (c) Number of terms in

1st series $\quad 817 = 17 + (m - 1)4$

$\Rightarrow \quad m = 201$

2nd series, $\quad 851 = 16 + (n - 1)5$

$\Rightarrow \quad n = 168$

Now, $17 + (m - 1)4 = 16 + (n - 1)5$

$\Rightarrow \quad 4m + 2 = 5n$

For, $m = 2$, $n = 2$

$m = 7, \; n = 6$

$m = 12, \; n = 10$

$m = 17, \; n = 14$

It also form a series.

$\therefore$ Number of common terms,

$$197 = 2 + (P - 1)5$$

$$\Rightarrow \quad P = 40$$

67. (b) (a) ONGEAR - ORANGE

(b) NOONI - ONION

(c) ALPEP - APPLE

(d) AUVAG - GUAVA

Onion is the only thing that is taken out of the soil.

68. (c) At C, number of passenger = 55

$\therefore$ Total number of passenger at B $= \frac{5}{4} \times (55 - 3)$

$$= \frac{5}{4} \times 52 = 65$$

$\therefore$ Total number of passengers at

$$A = \frac{6}{5} \times (65 - 10)$$

$$= \frac{6}{5} \times 55 = 66$$

69. (a) If the answer of question is 'No', then none belongs to No-zone. This implies both are Yes. Which is a contradiction.

If Kumar is 'Yes' the answer to the questions is 'Yes' that implies that Kevin is from No.

70. (a) Raman's DOB = 5th March, 1970

$\therefore$ Lakshman's DOB = 8th Feb 1970

Repulic Day, i.e. 26 Jan = Monday

$\therefore$ 8th Feb = Sunday.

71. (c) If P is true, then Q is true.

If Q is true, then R is true.

So, if P is true, then Q and R are both true.

and if Q and R are true, then S is false.

72. (b)

$E^- \Leftrightarrow A^+ \xleftrightarrow{\text{Brothers}} D^- \Leftrightarrow B^-$

$C^+ \quad F^- \qquad G^-$

So, C is son of A and E.

73. (b) Total hours from 5 am to 10 pm of 3rd day = 65 h

The clock looses 16 min in 24 h.

i.e. 23 h 44 min of the clock = 24 h of the correct clock

So, 65 h of clock $= \frac{24 \times 15 \times 65}{356}$ of the correct clock

$$= 65.73 \text{ h}$$

$$= 65 \text{ h } 44 \text{ min}$$

So, correct time = 10 : 45 pm (approx)

74. (b) 4 days after Monday is Friday

So, today is Tuesday.

Solutions. (Q. Nos. 75-77) From the given information the arrangement will be as follows.

Red Orange White
S U P

T Q R
Blue Green Yellow

Order of height T > S/ Q > P > R > U

75. (d) 2nd highest cannot be determined.

76. (b) 2nd shortest house is R.

77. (b) The tallest house is T which is Blue.

78. (a) NECESSARY

There is only one such pair NS.

79. (a) The series is as follows

4

7

$7 + 4 = 11$

$11 + 7 = 18$

$11 + 18 = 29$

$18 + 29 = 47$

$29 + 47 = \boxed{76}$

$47 + 76 = 123$

$76 + 123 = 199$

$\therefore$ Missing term = 76

Solutions (Q. Nos. 80-82) From the given information,

$A^+ \Leftrightarrow C^-$

$E^+ \Leftrightarrow B^-$

$D^+ \leftrightarrow F$

According to weight A>C>E>B>D,F

80. (d) BE is a couple

81. (a) C will be at 2nd place.

82. (a) C is grandmother of D.

Solutions (Q. Nos. 83-85) From the given information the arrangement will be as

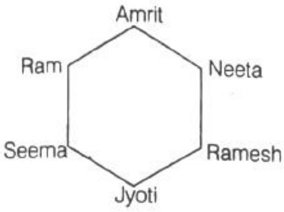
Face E K A J B I G C H D F

83. (d) I is sitting in the middle of the row.

84. (c) D and K will be at end points.

85. (b) CHDF are sitting to the right of G.

Sol. (Q.Nos. 86-89) From the given information the arrangement is as follows .



86. (a) After interchanging their position Jyoti will be opposite to Neeta.

87. (d) Ram is sitting left to Amrit.

88. (a) Neeta is sitting between Amrit and Ramesh.

89. (c) Amrit is sitting opposite to Jyoti.

90. (d) We can check it through option.

If x is the number of children in 1st row.
then in 2nd row $= (x-3)$ children
in 3rd row $= (x-6)$ children
....
....
6th row $= (x-15)$ children

$$\therefore \quad x + x - 3 + x - 6 + x - 9 + x - 12 + x - 15 = 630$$
$$\Rightarrow \quad 6x - 45 = 630$$
$$\Rightarrow \quad 6x = 675$$
$$\Rightarrow \quad n = \frac{675}{6} = 112.5$$

So, 6 row is not possible as number of children cannot be in fraction.

General English

91. (b) In the passage, "basic" has the context of being important. So, its opposite word will be "unimportant'.

92. (d) The reason for the cement prices going-up is not given in any of the statements.

93. (a) The crisis faced by the cement tile manufactures is the very high prices of white cement.

94. (a) 'Artificial' generally gives the sense of 'man-made' but in the passage it is used to denote the sense of 'deliberate'.

95. (d) A person who makes money by starting or running business-entrepreneur.

96. (b) Group of insects is denoted by the phrase-swarm of flies.

97. (c) 'Reputation' and 'fame' are the similar words.

98. (b) 'Theatrical' and 'dramatic' are similar from the options, 'histrionic' gives the same sense.

99. (a) Uncertain, dubious and doubtful have the same meaning. Only "indisputable" is different from the group.

100. (c) Scale : Tone gives the sense of "skin" and its colour" with the same theme the only spectrum : colour is similar.

101. (d) Idiom "to burn a hole in the pocket means" money that is spent quickly.

102. (b) Phrasal verb "look out on" means "intended in a certain direction. Hence" out on" will be the correct choice.

103. (c) The adjective "dark" has been used in the superlative degree. So, article "the" is suitable usage.

104. (a) "steak" means "a piece of meat". So, the item is "edible". Hence, (a) is the suitable usage.

105. (b) The correct phrase will be "for ages as it denotes a period of time.

106. (b) "Should" will be followed by either. "be" or "have been. But "be is not appropriate as "hour" is not preceded by "at". Hence "have been" is the suitable usage.

107. (d) As the statement is negative the tag must be positive. The auxiliary verb is 'should'. So, the tag will be 'should you'.

108. (a) Mathematics was studied by them last year.
Structure : obj + was/were + V_3 + by + sub. (obj. case) + complement.

109. (b) 'Egress' means 'exit'.

110. (a) and (c) both.
Idiom 'elbow room' has two senses.
First : opportunity for freedom of action.
Second : to give enough space to move or work in. Hence, both the options are correct.

Computer Awareness

111. (a) 42797 KB will take 85512 sectors (42797*1024 bytes/512 bytes).
Since, there are 64 sectors per surface 85512/64 = 1337.406 sectors are required, so we take 1338 sectors. These sectors are distributed among 16 surfaces, so 1338/16 = 83.58 cylinders will be required. So, the final answer will be 84 + 1200 = 1284.

One more fact to be noted is that the file occupies 83.58 cylinders, but the 0.58 cannot be accommodated in the first one (the files storage starts from < 1200, 9, 40 >).
Hence, the file will be extended to 194 (85594 – 85400) more bytes of cylinder 1284.

112. (b) K-map for $F(P, Q, R, S) = \Sigma(0, 2, 5, 7, 8, 10, 13, 15)$

where, 2, 7, 8, 13 are do not care terms.

$PQ \backslash RS$	$\bar{R}\bar{S}$	$\bar{R}S$	RS	$R\bar{S}$
$\bar{P}\bar{Q}$	1 (0)	(1)	(3)	X (2)
$\bar{P}Q$	(4)	1 (5)	X (7)	(6)
PQ	(12)	X (13)	1 (15)	(14)
$P\bar{Q}$	X (8)	(9)	(11)	1 (10)

There are 2 quads:

Quad 1 $(m_0 + m_2 + m_8 + m_{10}) = \bar{Q}\,\bar{S}$

Quad 2 $(m_5 + m_7 + m_{13} + m_{15}) = QS$

Hence, minimal SOP form $= QS + \bar{Q}\,\bar{S}$

113. (c) The given Venn diagram represent XNOR expression.

Hence, a XNOR $b = ab + a'b'$

114. (b) The range of values for n-bit signed magnitude representation is $-(2^{n-1} - 1)$ to $(2^{n-1} - 1)$.

115. (a) We know that,

$$\text{Average latency} = \frac{\text{Max latency}}{2}$$

$$\text{where, Max latency} = \frac{1}{\text{rev / time}} = \frac{1}{\frac{6000}{60}}\text{ s}$$

$$= \frac{60}{6000}\text{ s} = \frac{1}{100}\text{ s}$$

$$= 10^{-2}\text{ s}$$

$$\text{So, the average latency} = \frac{10^{-2}}{2}$$

$$= 0.5 \times 10^{-2}$$

$$= 5 \times 10^{-3}\text{ s}$$

116. (d) 8 bit representation of $100 \rightarrow 01100100$

Complement $\rightarrow 10011011$

add $1 \rightarrow 00000001$

$-100 \rightarrow 10011100$

Hence, 2's complement representation is 10011100.

117. (*) None option is correct.

118. (b)

119. (c) Given, $(43)_x = (y3)_8$

By converting to decimal system

$$4x^1 + 3x^0 = y \cdot 8^1 + 3 8^0$$

$$4x + 3 = 8y + 3$$

$$4x = 8y$$

$$x = 2y$$

x is the base of number system $x > 4$

y is number in the number system with base 8, so $y < 8$

x	y
6	3
8	4
10	5
12	6
14	7

Number of possible solutions = 5

120. (d) $(1101)_2 - (1010)_2$

1's complement of 1010 is 0101.

Minued — 1 1 0 1

1's complement of subtrahend — 0 1 0 1

Carry over 1 0 0 1 0

1

0 0 1 1

The required difference is 0011.